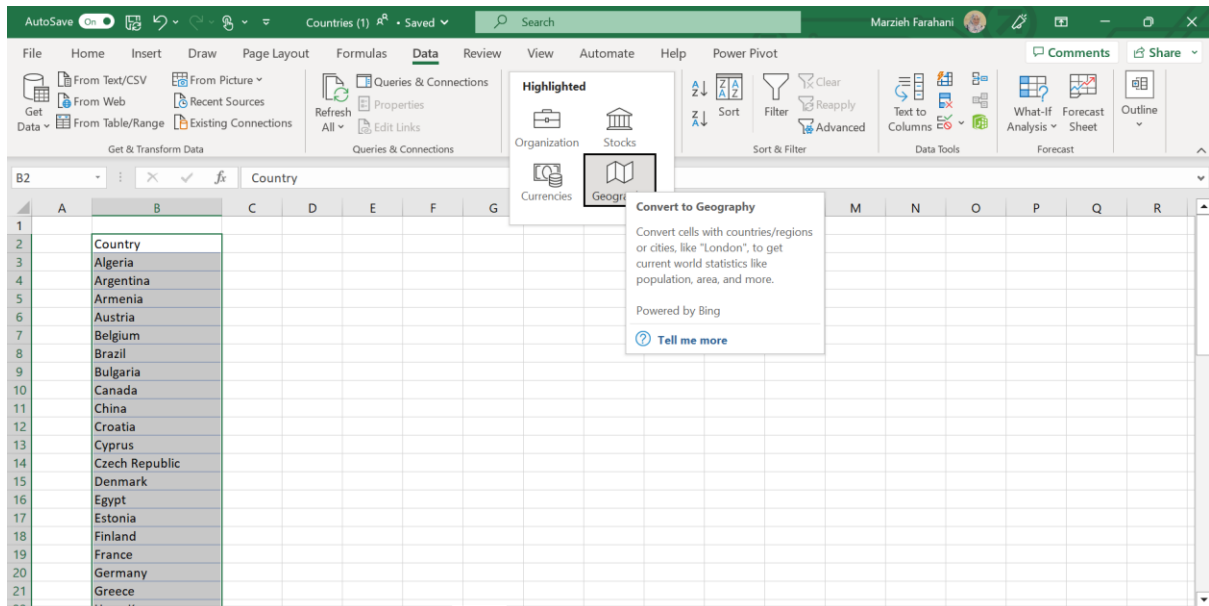


How to get geographic data in Excel

To insert geographic data into Excel, you first convert text into the Geography data type.

1. Download all countries name from Moodle.
2. Select all countries cells.
3. With the cells still selected, go to the Data tab, and then click Geography.



4. Excel will convert your text to the Geography data type. They are converted if they have this icon:
5. Select all cells and the **Insert Data button** will appear. Click that button, and then click a field name to insert data from the world bank.
6. Pick fields: Area, Carbon dioxide emissions, CPI Change (%), Electric power consumption, Fossil fuel energy consumption, Gasoline price, GDP, Unemployment rate, Minimum Wage, Population
7. Type all field names in the header row. Click on each piece of data and see what data has been inserted.

Get & Transform Data				Queries & Connections		Data	
E2				=B2.[Carbon dioxide emissions]			
	C	D	E	F	G	H	
1	Population	Area	Carbon dioxide emissions	CPI	Electric power consumption	Fossil fuel energy consumption	
2	43,053,054	2,381,741	150,006	151.36	1,363	10	
3	44,938,712	2,780,400	201,348	232.75	3,075	8	
4	2,957,731	29,743	5,156	129.18	1,962	7	
5	8,877,067	83,871	61,448	118.06	8,356	6	
6	11,484,055	30,528	96,889	117.11	7,709	7	
7	212,559,417	8,515,770	462,299	167.40	2,620	5	

8. Although it's not required, we recommend creating an Excel table. To create a table, go to **Insert > Table**.
9. Add a new calculated field to your data. (GDP per capita = GDP/Population)

=G2/K2							
E	F	G	H	I	J	K	L
fuel energy consumption	Gasoline price	GDP	Unemployment rate	CPI Change (%)	Minimum Wage	Population	GDP per Capita
100.0	\$ 0.28	\$ 169,988,236,398	11.70%	2.0%	\$ 0.95	43,053,054	=G2/K2
87.7	\$ 1.10	\$ 449,663,446,954	9.79%	53.5%	\$ 3.35	44,938,712	10006.14897
74.6	\$ 0.77	\$ 13,672,802,158	16.99%	1.4%	\$ 0.66	2,957,731	4622.733493
65.7	\$ 1.20	\$ 446,314,739,528	4.67%	1.5%	#FIELD!	8,877,067	50277.27509
75.9	\$ 1.43	\$ 529,606,710,418	5.59%	1.4%	\$ 10.31	11,484,055	46116.69923
59.1	\$ 1.02	\$ 1,839,758,040,766	12.08%	3.7%	\$ 1.53	212,559,417	8655.264804

10. Gross domestic product per capita is sometimes used to describe the standard of living of a population, with a higher GDP meaning a **higher standard of living**.



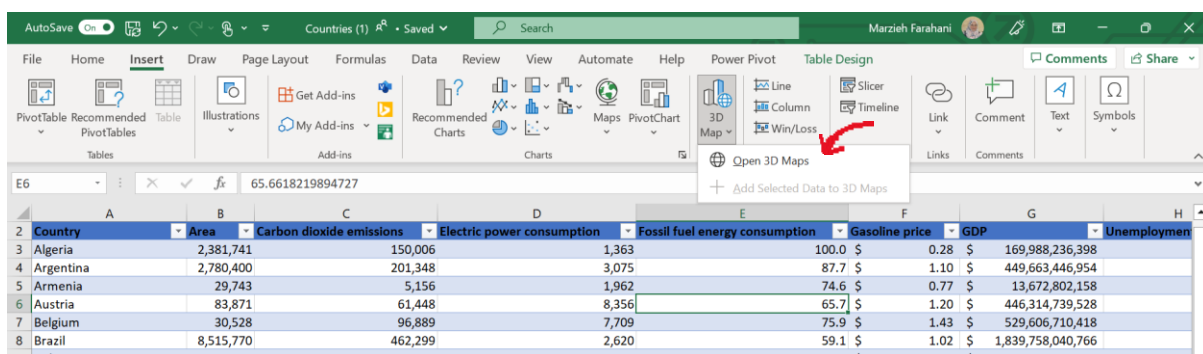
$$\text{GDP Per Capita Formula} = \frac{\text{GDP of the Country}}{\text{Population of that Country}}$$



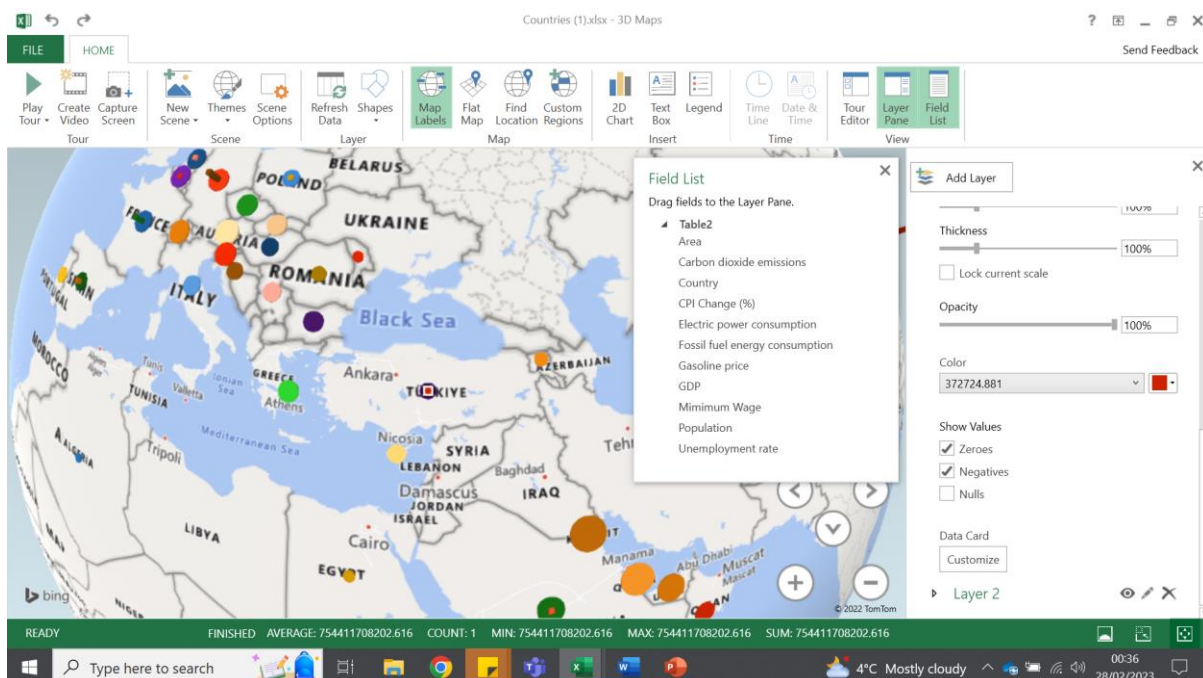
11. Use pivot table and Pivot chart to show top ten countries with highest Electric power consumption, Fossil fuel energy consumption, Gasoline price, GDP per capita, Unemployment rate, Minimum Wage, Population and CO2 emission.

How to Create a 3D Chart Using Microsoft 3D Maps in Excel

1. Open the Excel workbook containing the geographic data sets you want to visualize in Microsoft 3D Maps.
2. Now, click on any of the cells within the data sets.
3. Click on the Insert option in the Ribbon to open a menu that shows elements like Tables, Charts, Sparklines, etc.
4. On the extreme right-hand side, you will see 3D Map within the Tours section of the Ribbon.
5. Click on the dropdown menu of the 3D Map button to find the Open 3D Maps option.
6. Click on the Open 3D Maps to activate Microsoft 3D Maps in Excel for the first time.



	A	B	C	D	E	F	G	H
	Country	Area	Carbon dioxide emissions	Electric power consumption	Fossil fuel energy consumption	Gasoline price	GDP	Unemployment
3	Algeria	2,381,741	150,006	1,363	100.0	\$ 0.28	\$ 169,988,236,398	
4	Argentina	2,780,400	201,348	3,075	87.7	\$ 1.10	\$ 449,663,446,954	
5	Armenia	29,743	5,156	1,962	74.6	\$ 0.77	\$ 13,672,802,158	
6	Austria	83,871	61,448	8,356	65.7	\$ 1.20	\$ 446,314,739,528	
7	Belgium	30,528	96,889	7,709	75.9	\$ 1.43	\$ 529,606,710,418	
8	Brazil	8,515,770	462,299	2,620	59.1	\$ 1.02	\$ 1,839,758,040,766	



Take screenshots from your map and add to your countries data analysis.

Create your Excel dashboard (go to Step 6)

<https://clickup.com/blog/how-to-create-a-dashboard-in-excel/>